

JalNaadi

Master Explainer

From the problem to the working idea — written for a reader with no background in water supply, signal processing or electronics.

JalNaadi in 60 seconds. Villages across India now have tap water (about 15.83 of 19.36 crore rural households, PIB, 31 Mar 2026). The pipes are buried, they leak, and nobody can see where. The sensors Jal Jeevan Mission already uses can tell that a *village* is losing water — not *which pipe*. JalNaadi (“pulse of the pipe”) is a low-cost system of clamp-on sensors that **(1)** notices the loss from the night-time water flow, **(2)** wakes pairs of vibration sensors that listen to the leak, and **(3)** works out *where* the leak is from the tiny time difference between the two sensors — then shows it on a map and ranks repairs. Tagline: **Listen first, locate next.**

Team: SolveX TSDCEM (Team ID 128664) · **Problem statement:** SIH26254 — Development of Indigenous Leak Detection Sensor Systems for Operation & Maintenance of Water Supply Pipelines · **Document version:** 1.0, 4 October 2026

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1 The problem

1.1 How water reaches a village tap

Water is drawn from a source (a borewell, river or dam), treated, and pumped into a tank raised on legs — the **service reservoir (SR / ESR / OHT)**. From the tank, gravity pushes it through a network of buried pipes to every house, where it comes out of a **functional household tap connection (FHTC)**. The village committee (**VWSC / Gram Panchayat**) runs the scheme day to day, with engineers of the state's **PHED / Rural Water Supply** department above it. The government norm is **55 litres per person per day (lpcd)**.

Under the **Jal Jeevan Mission (JJM)**, about **15.83 crore of 19.36 crore rural households** now have such a tap (PIB, 31 Mar 2026). JJM 2.0, approved by the Cabinet on 10 March 2026, shifts the emphasis from laying pipes to **keeping the service working** — operation and maintenance (O&M).

Term	Meaning
SR / ESR / OHT	The raised tank that feeds the village by gravity
DMA	District Metered Area — a zone of the network with a metered inlet
NRW	Non-revenue water: water put into the system that never reaches a paying/consuming tap (leaks are the biggest part)
FHTC	Functional Household Tap Connection
VWSC	Village Water and Sanitation Committee
PHED / RWS	State engineering department for water supply
lpcd	Litres per capita per day

1.2 What goes wrong

Pipes crack, joints loosen, valves wear, roots and ground movement stress them. Water escapes into the soil. Because the pipe is buried, **the leak is invisible**. What the village sees is low pressure at the tail end, an empty tap, or a damp patch that appears long after the damage began. Meanwhile the lost water has already been treated and pumped — wasted money and electricity.

Nobody knows **where** to dig. Crews either wait for a road to collapse, or dig trial holes along the pipe. A small leak can run for months. Leaks also lower pressure, which can let dirty water seep *into* the pipe (ingress) — an indirect health risk.

1.3 Where the government's own monitoring stops

The NJJM Technical Expert Committee Report on measurement and monitoring (March 2021) sets out sensor-based monitoring for rural schemes. Its “intermediate design” puts a **bulk flow meter at the tank outlet**, flow meters at zone (DMA) entries and a **pressure sensor at the tail end**, reporting over **LoRa in the 865-867 MHz band**. The report itself lists frequent leakages and the difficulty of detecting minor leakages as a problem. These meters give the *water balance* of a zone: they can show that more water goes in than comes out. They cannot say **where inside the zone** the pipe is broken.

The gap in one line: JJM sensors tell **THAT** a village is losing water. JalNaadi tells **WHERE**.

2 The problem statement (SIH26254)

The Ministry of Jal Shakti (NJJM, Department of Drinking Water and Sanitation) asks for **indigenous leak-detection sensor systems** for the operation and maintenance of water-supply pipelines. Our deck maps JalNaadi to five expectations in the brief:

What the problem statement expects	How JalNaadi answers
Detect and accurately locate leaks	GATE flags a leak-suspect zone; an EAR pair pinpoints the position along the pipe
Real-time alerts and monitoring	LoRa → gateway → MQTT → dashboard and SMS; 24×7 node health, event-triggered acoustic sensing

What the problem statement expects	How JalNaadi answers
Energy-efficient and cost-effective	Sensors sleep and wake only on a GATE flag; about 16 mAh per day; node BOM $\approx$ ₹3,000–5,500 (indicative)
Work with existing infrastructure	Reuses JJM flow meters; clamp-on at valves and hydrants; NJJM-style JSON; Sujal Gaon ID tagging
Remote monitoring and maintenance planning	Priority-ranked repair list; node and battery health; over-the-air configuration

3 What exists, and the gap

Approach	Detects a leak	Locates the pipe segment	Remote always-on	Skill needed	Cost at rural scale
Flow-meter water balance (existing JJM IoT)	Yes — zone level	No	Yes	Low	Low (already deployed)
Manual ground-microphone survey	Yes — at the point listened to	By ear, by a skilled person	No	High	Low per tool, high per village visit
Industrial leak-noise correlator	Yes	Yes	Mostly survey-based	Trained crew	High (the brief itself calls current technologies expensive)
JalNaadi	Yes	Yes — design goal ± 1 m	Yes (LoRa)	Low, app-guided	Low (design target)

4 The physics, in plain words

Three ordinary facts make the whole system possible.

Fact 1 — A leak makes noise

Water squirting out of a pressurised pipe through a small hole produces a steady hiss. That hiss is a vibration that travels along the pipe wall and the water inside. A small **accelerometer** (the same kind of chip that is in a phone) clamped to the pipe can feel it. The further you are from the leak, the weaker it gets — plastic pipes weaken it faster than metal ones.

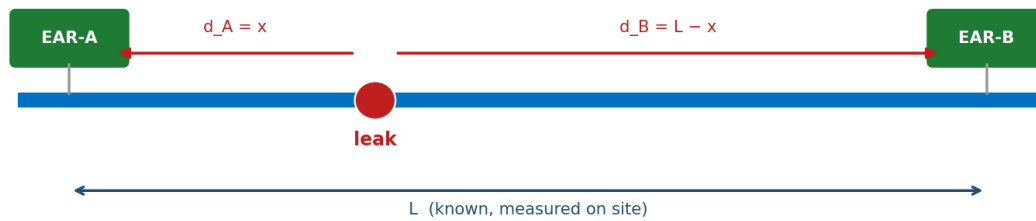
Fact 2 — The night is quiet

Between about 2 and 4 AM almost nobody uses water, and there is little traffic and machinery noise. So (a) whatever flow still enters a zone is mostly leakage, and (b) the leak's hiss stands out best against the background. Leak surveyors have long worked at night for the same reason. JalNaadi uses a **02:00–04:00 listening window** (adjustable per village).

Fact 3 — Two sensors can find the spot

Put one sensor **A** and one sensor **B** on the pipe, on either side of the leak, a known distance **L** apart. The hiss reaches the nearer sensor first. The **time gap τ** between the two arrivals tells you where the leak is — like estimating where a clap happened from the delay between two microphones.

$$\tau = t_B - t_A = (d_B - d_A) / v = (L - 2x) / v \Rightarrow x = (L - v \cdot \tau) / 2$$



Example: $L = 20$ m, $v = 400$ m/s, leak 6 m from A $\rightarrow t_A = 15$ ms, $t_B = 35$ ms, $\tau = 20$ ms $\rightarrow x = (20 - 8)/2 = 6$ m

Figure 1 — Locating a leak from the arrival-time difference between two sensors.

The worked example in Figure 1: A and B are 20 m apart, the leak is 6 m from A, sound travels at 400 m/s. A hears it after $6/400 = 15$ ms and B after $14/400 = 35$ ms. The gap is $\tau = 20$ ms, and $x = (L - v \cdot \tau)/2 = (20 - 8)/2 = 6$ m. (400 m/s is a typical figure for water-filled plastic pipe; the real value differs by pipe material — see below.)

Two practical problems, and our answers

- **The speed v is not known.** It depends on pipe material, diameter and age. Using a wrong v shifts the answer — by metres on a long segment. **One-Tap Commissioning:** at installation a worker taps the pipe beside A with a spanner. Both sensors record the tap, the A $\rightarrow$ B arrival gap Δt is measured, and $v = L/\Delta t$ is stored for that segment. No datasheet needed, and it works on mixed HDPE / PVC / GI / DI pipes.
- **The two sensors must share a clock.** A timing error δt becomes a position error of about $v \cdot \delta t / 2$ (1 ms at 400 m/s ≈ 0.2 m). Instead of trying to keep two clocks identical for hours, the gateway sends one radio **beacon**; both nodes record the instant it arrives (the same instant for both, because radio waves cover these distances in microseconds), and start recording a fixed time later. A second beacon at the end of the burst removes any crystal drift. This is called reference-broadcast synchronisation.

5 Our solution: GATE $\rightarrow$ EAR $\rightarrow$ DESK

JalNaadi has three layers. The cheap hydraulic layer decides *when* the more expensive acoustic layer runs.

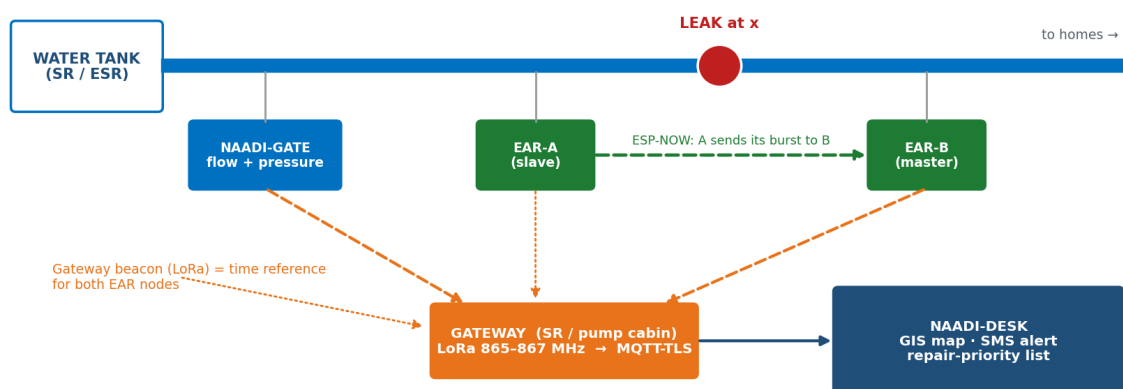


Figure 2 — System architecture on a village pipeline.

Layer	What it does	How
NAADI-GATE (screen)	Decides <i>which zone</i> probably has a leak	Takes the zone's inlet flow (from the existing JJM bulk flow meter, or our own flow + pressure node), averages the 02:00-04:00 window to one number per night, compares it with the zone's learned night baseline (EWMA), and accumulates deviations with a CUSUM change detector. A sustained rise = "leak suspected in zone Z". The baseline freezes while a change builds, so a leak cannot be learned away.

Layer	What it does	How
NAADI-EAR (locate)	Finds <i>where</i> in that zone	Clamp-on vibration nodes at valves / hydrants sleep until the gateway's beacon arrives. Both nodes of a pair record a synchronised 10-second burst, classify it on the node (leak hiss vs pump hum vs tap knock), the slave sends its burst to the master over ESP-NOW, and the master cross-correlates (GCC-PHAT) to get τ , then $x = (L - v \cdot \tau)/2$.
NAADI-DESK (act)	Turns a number into a repair	A LoRa gateway at the tank or pump cabin forwards compact JSON over MQTT-TLS to a server. The dashboard pins the leak on a map, sends an SMS, and ranks repairs by estimated litres lost per day $\times$ households served.

A night in the life of the system

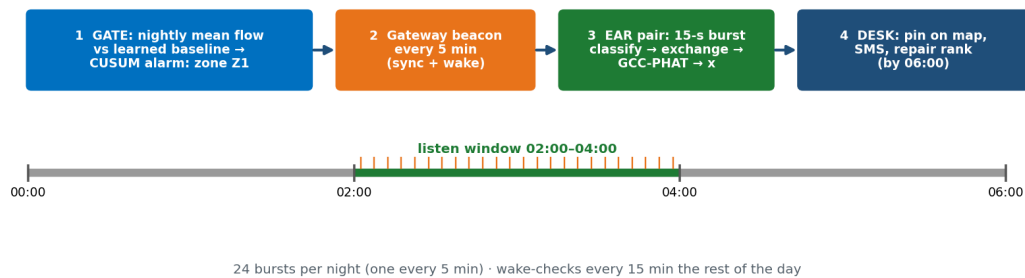


Figure 3 — From the 02:00 screening to a repair ranking by morning.

- 02:00 — GATE has already flagged zone Z1 from the previous nights' flow. The gateway starts sending a beacon every 5 minutes.
- Each beacon wakes the EAR nodes in Z1 (they check for it every 15 minutes), synchronises them, and starts a 15-second burst (10 s recording + about 5 s of processing and exchange).
- Bursts that sound like a leak are exchanged inside the pair; bursts that sound like a pump or a tap stop there. The pair master reports a position, e.g. “segment S3, 6.0 m from valve A, confidence 31”.
- By 06:00 the pump operator's phone shows the pin and the ranked work order.

6 Inside a node

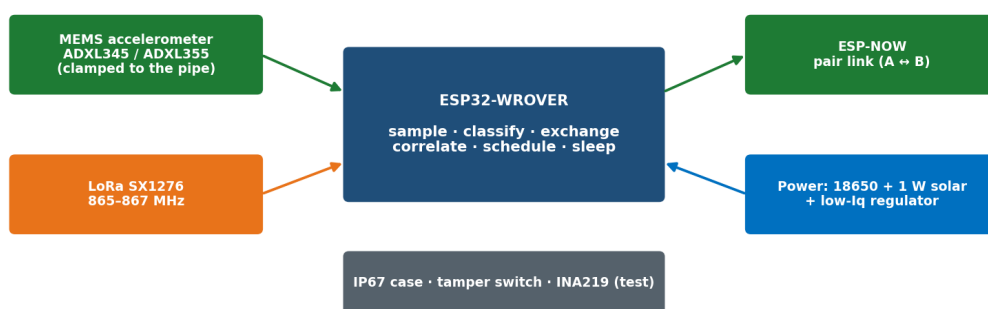


Figure 4 — One EAR node.

Block	Part (prototype)	Why
Brain	ESP32-WROVER-E, 8 MB PSRAM	Low-power dual-core MCU with hardware timer (μ s) and deep sleep; PSRAM holds the partner's 64 kB burst
Sensor	ADXL345 (3.2 kHz) — or ADXL355 (4 kHz, low-noise) for field units	MEMS accelerometer clamped to the pipe; both nodes of a pair must use the same sensor and sampling rate
Long-range radio	SX1276 LoRa, 866 MHz	Licence-exempt 865–867 MHz band in India; only short result messages travel on it

Block	Part (prototype)	Why
Pair link	ESP-NOW (built into the ESP32)	Moves the 64 kB burst from node A to node B in about a second
Power	18650 Li-ion + 1 W solar + charger, low-quiescent regulator	Solar-recharged, designed to run unattended
Housing	IP67 case, clamp/saddle with rubber pad, tamper switch	Weatherproof, tamper-evident, no pipe cutting

Power budget (calculated, per node, per day)

Item	Working	mAh/day
Deep sleep	24 h × 0.15 mA	3.6
Listening bursts	24 bursts × 15 s × 100 mA (one burst every 5 min, 02:00–04:00)	10.0
Wake-checks	96 × 1 s × 50 mA (every 15 min)	1.3
LoRa uplinks	24 × 1 s × 120 mA	0.8
Total		≈ 15.7 ≈ 16
Battery-only autonomy	2,500 mAh ÷ 16 mAh/day	≈ 156 days
Solar margin	1 W × 4 sun-hours × 50 % loss = 2 Wh ≈ 540 mAh/day at 3.7 V	> 30 × the load

The 0.15 mA sleep figure needs a low-quiescent-current board (no USB-UART chip, efficient regulator). A stock development board draws far more in sleep. The design KPI is **≥ 120 days on battery alone** (derated for cold and ageing); the power budget is verified on the rig with an INA219 logger.

Bill of materials (indicative prices, to be finalised from actual purchase)

Item	Indicative ₹
ESP32-WROVER-E board	650
SX1276 LoRa module + antenna	700
ADXL345 breakout (ADXL355: about ₹3,000)	300
18650 cell + holder	350
Charger / regulator	120
1 W solar panel	300
IP67 case + gland	300
Clamp, magnet, pad	150
Wiring, connectors	150
Node total	≈ 3,000 with ADXL345; ≈ 5,500 with ADXL355 (the deck quotes ₹3,000–5,500)

Prices are indicative; replace them with real quotes before quoting a final figure. A localisation pair plus gateway therefore costs about ₹8,000–13,000.

7 How a leak is located — the nine steps

#	Step	What happens
1	Beacon	Gateway transmits a LoRa packet. Every node that hears it records the time of arrival with its microsecond hardware timer.
2	Synchronised start	Each node starts its accelerometer a fixed 200 ms after the beacon, so both record the same instants (to well under 1 ms).
3	Record	10 seconds at 3.2–4 kHz, one axis, 16-bit samples $\approx$ 64 kB per node.
4	Classify on the node	Rule-based features: in-band energy fraction (50–1,000 Hz), local spectral-line prominence (pump hum is a narrow line), kurtosis (a tap is a spike). Leak = steady broadband hiss.
5	Exchange	Only if the burst looks like a leak, node A sends its record to node B over ESP-NOW (about 1 s).
6	Correlate	Node B whitens both signals in the 50–1,000 Hz band (GCC-PHAT), averages over 2-second segments, and finds the lag where they match best, refined below one sample by parabolic interpolation.
7	Convert	$x = (L - v \cdot \tau)/2$, with v from One-Tap Commissioning.
8	Quality check	Peak-to-background ratio gives a confidence number. Low confidence is flagged, not hidden.
9	Report	About 110 bytes of JSON go out over LoRa: segment, x , L , v , confidence, class.

Why LoRa never carries the raw data

A 10-second burst is $3,200 \text{ samples/s} \times 10 \text{ s} \times 2 \text{ bytes} = \mathbf{64 \text{ kB}}$. LoRa at spreading factor 7 and 125 kHz bandwidth moves about 5.5 kbit/s — roughly **95 seconds** of airtime per node per burst (about 5 minutes at SF9). With 24 bursts a night that would flatten the battery. So the architecture keeps the heavy data **inside the pair**: ESP-NOW moves 64 kB in about a second over distances of tens to a couple of hundred metres, which is exactly the range over which two sensors can usefully hear the same leak. Only the 110-byte result goes by LoRa (about 0.6 s of airtime at SF9).

Memory is the other constraint: node B holds its own burst, A's burst and the FFT buffers — about 130 kB plus working space, which is why the prototype uses the PSRAM variant of the ESP32. If the pair link is ever stressed, our simulation (Part 9) shows 8-bit samples (half the data) and 4-bit samples (a quarter) work as well as 16-bit, and even 1-bit signs (one sixteenth) matched 16-bit in simulation at normal signal quality.

8 Software and data

Night-flow screening (GATE)

Each night the zone's mean inlet flow over 02:00–04:00 is one number (m^3/h). The first 14 nights teach a baseline and its normal spread. After that the baseline follows slow, legitimate drift, but freezes while a change is building. Each night's deviation is standardised and added to a one-sided CUSUM score (allowance $k = 0.75$, threshold $h = 5$). When the score crosses the threshold the zone is flagged. The size of the persistent rise is the **estimated leak flow**.

Repair priority (DESK)

Estimated loss in litres per day = excess flow (m^3/h) $\times 1,000 \times 24 \times 0.8$, where 0.8 corrects for night pressure being higher than the daily average. **Priority score = loss (L/day) $\times$ households served**. Example: 0.5 m^3/h on 100 households scores $9,600 \times 100 = 960,000$; 0.1 m^3/h on 400 households scores $1,920 \times 400 = 768,000$, so the first is repaired first.

Data format

Messages follow the field layout of the NJJM reference payload (gid, did, dty, dt, p); field codes should be aligned with the latest NJJM specification before field use:

```
{"gid": "G01", "did": "E01", "dty": "ear", "dt": 1759533600, "p": {"seg": "S3", "x": 6.0, "L": 20.0, "v": 400, "cf": 31.2, "cls": "leak"}}
```

The backend (Python FastAPI + SQLite, MQTT in) stores nightly flows and located leaks, runs the GATE detector per zone, converts x along a segment into map coordinates, and returns the ranked list; the dashboard (Leaflet map) shows pins and the work list. All of this is in the repository (Appendix C) and its tests pass.

9 Simulation results — what our own experiments show

Before building hardware we simulated the signal chain in Python (repository folder analysis/, fully reproducible). **These are simulations, not hardware measurements.** They assume a single, non-dispersive wave, independent white sensor noise and no reflections; real pipes add dispersion and echoes, which the rig tests will measure. What the simulation is good for is showing *which factors matter*.

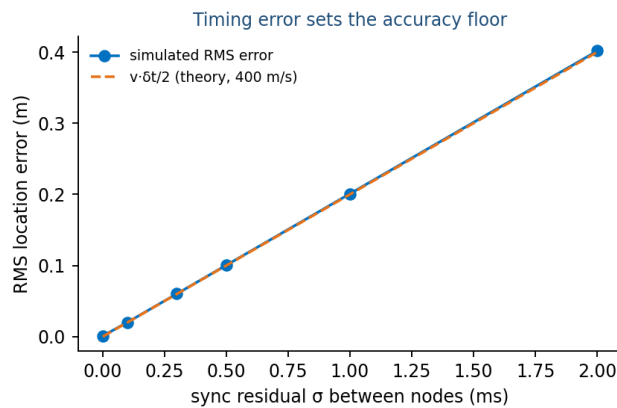


Figure 5 — With a clean signal, timing is what limits accuracy: error $\approx v \cdot \delta t / 2$.

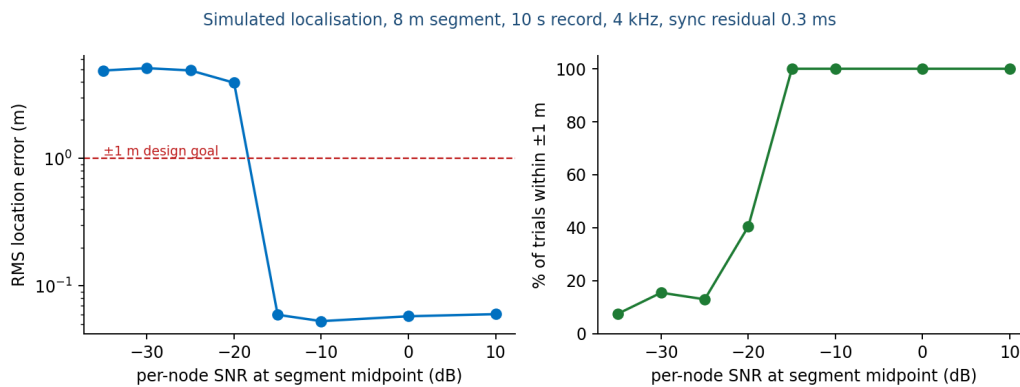


Figure 6 — Correlation over 10 s works down to very poor signal-to-noise ratio (idealised noise); accuracy stays at the timing floor until the signal drowns.

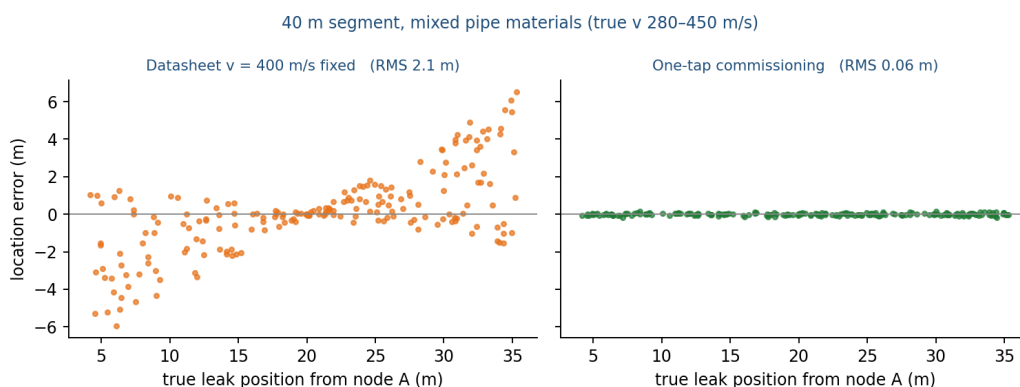


Figure 7 — One-Tap Commissioning: on a 40 m segment with unknown pipe material, a fixed datasheet speed gives 2.1 m RMS error (55% within ± 1 m); calibrated, 0.06 m.

Key takeaways

- **Timing dominates.** RMS error is 0.02 m at 0.1 ms sync residual, 0.06 m at 0.3 ms, and 0.20 m at 1 ms — matching $v \cdot \delta t / 2$. Sub-millisecond beacon sync is enough for sub-metre location in this model.
- **Wave-speed calibration is the biggest single accuracy lever.** Without it, errors of several metres appear on long segments; with One-Tap Commissioning they fall below 0.1 m in simulation.
- **Averaging over 10 s** gives huge processing gain: the correlation peak survives per-node SNR down to about -15 dB in the idealised model and breaks down near -20 dB.

- **Compression is safe:** at a hard SNR of -17 dB, 8-bit and 4-bit samples keep 96% / 95% of results within ± 1 m (16-bit: 95.5%); 1-bit drops to 64% at that extreme, but matched 16-bit at 0 dB and -10 dB.

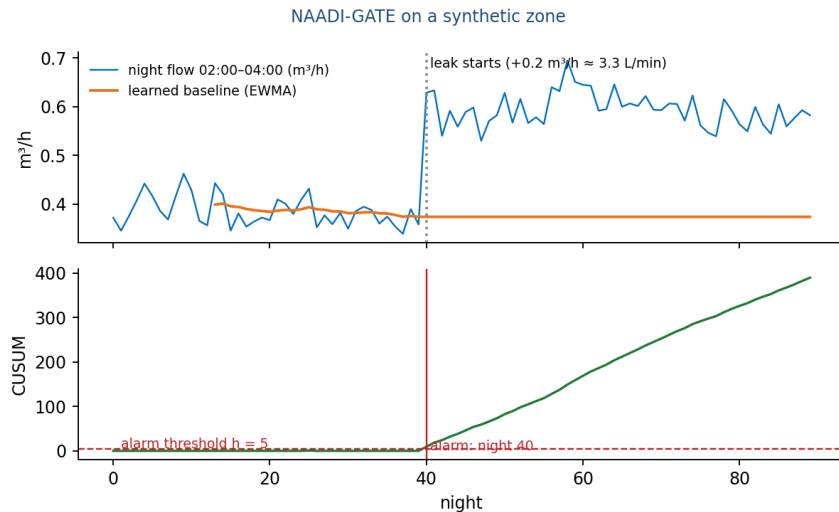


Figure 8 — NAADI-GATE on a synthetic zone: baseline, CUSUM and the alarm.

Leak size	L/min	L/day	Detected	Median nights to alarm	90th percentile
0.1 m ³ /h	1.7	2,400	100%	2.0	5.0
0.2 m ³ /h	3.3	4,800	100%	0.0	1.0
0.3 m ³ /h	5.0	7,200	100%	0.0	0.0
0.5 m ³ /h	8.3	12,000	100%	0.0	0.0

Synthetic zone: baseline night flow 0.4 m³/h with day-to-day variation 0.035 m³/h. False-alarm rate: about one per 467 nights. For scale, 0.1 m³/h is 2,400 L/day — under 2 % of a 500-household village's supply.

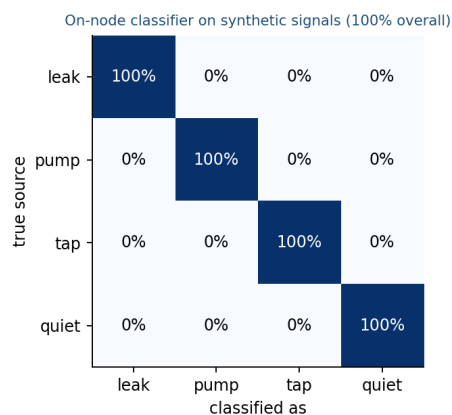


Figure 9 — On-node classifier on synthetic signals.

Honest reading. The classifier separates the four synthetic classes perfectly, which only shows the features are sensible — synthetic classes are easy. Its real accuracy has to be measured on recordings from the rig and, later, a pilot village. Likewise the location accuracy above is an *upper bound* set by timing and calibration; the rig will show how much real propagation costs.

10 Prototype plan

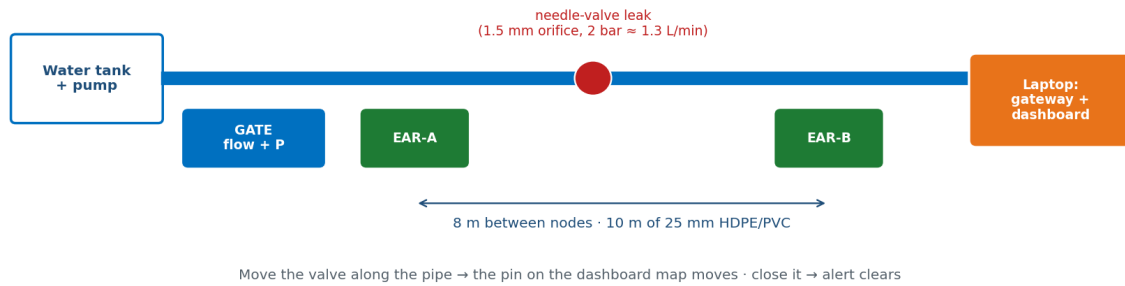


Figure 10 — Lab rig: a pump circulates water through 10 m of pipe; a needle valve makes a controllable leak.

The rig is deliberately small and cheap — a corridor, a bucket and a pump. A 1.5 mm orifice at 2 bar leaks about 1.3 L/min (orifice equation, discharge coefficient 0.6), which defines the “1 L/min” detection KPI.

Test	Variable	What we record	Design KPI
Localisation	Valve position along the pipe (8 positions × 5 repeats); HDPE and PVC	Reported x vs tape-measured x	±1 m on an 8 m segment
Calibration	Tap test on each pipe	v from tap vs v from a known leak position	v within 3 %
Sensitivity	Orifice 0.8 / 1.0 / 1.5 mm; pressure 1–3 bar	Detection yes/no and confidence	≥ 1 L/min detected
False alarms	Pump on, tap drawn, knocking, quiet	Classifier output per 4-s window	≤ 1 per node per week (extrapolated)
Synchronisation	Beacon sync residual (tap test at the middle)	τ at the centre should be 0	< 1 ms
Power	Full night cycle with INA219 logging	mAh per day	≈ 16 mAh/day
Dashboard	End-to-end: valve move → pin moves → SMS	Latency, correct segment	Works live in the demo

Suggested roles for a team of six

Role	Owens
Integration lead	Architecture, schedule, final demo, judge Q&A;
Firmware A	ADXL345 sampling, beacon sync, deep-sleep cycle, power logging
Firmware / DSP B	ESP-NOW exchange, on-device GCC-PHAT (ESP-DSP), classifier thresholds
Backend + dashboard	MQTT, FastAPI, map, SMS, ranking
Hardware + rig	Pipe rig, pump, valve, clamps, enclosure, wiring, prototype photos
Documentation + media	README, demo video, test log, slides for later rounds

Suggested build order (3 phases)

- **Phase 1 — proof of principle (first):** two nodes + a laptop, wired time reference if needed, record a leak on the rig, run gcc_phat offline and get a position. This single result is the most valuable thing the team can have.
- **Phase 2 — the real chain:** beacon sync, on-node classifier, ESP-NOW exchange, on-device correlation, LoRa uplink to the gateway.
- **Phase 3 — the story:** backend, dashboard map, SMS, GATE node, demo video, prototype photographs.

11 Risks and how they are handled

Risk	Handling
Leak noise fades quickly in plastic pipes	Shorter node spacing, set per material from rig tests; nodes only at valves and hydrants
Time-sync error between nodes	Beacon sync at burst start and end, drift correction, low-confidence flag
Pump, traffic and tap noise	Night window, on-node classifier, both-node agreement before alert
No public leak-acoustic dataset for Indian pipes	Record our own labelled rig data; synthetic injection; request pilot data from State PHED / NJJM
Raw data volume vs LoRa capacity	Pair link over ESP-NOW; only a ~110 B result on LoRa; 8-bit samples if needed
ESP32 memory	PSRAM variant, 2-second segment processing
Weather, dust, vandalism, battery shortfall	IP67 case, tamper switch, INA219 telemetry, install inside tank / valve chambers, solar margin > 30×
Adoption by small committees	App-guided commissioning, local-language alerts, ranked work list, training via VWSC / ITI

12 Impact — the arithmetic

A village of 500 households × 5 persons × 55 lpcd = **137,500 L/day**. If 10 % leaks away: **13,750 L/day ≈ 5.0 million L/year**, the daily water for 250 people. If JalNaadi halves that loss: about **2.5 million L/year per village** (about 2.5 billion L/year across 1,000 villages with the same assumptions). Household size and the 10 % leakage figure are assumptions; 55 lpcd is the JJM service norm. Beyond water: fewer outages, targeted digging instead of blind excavation, saved pumping energy, local IoT-technician jobs, and a path toward the AMRUT 2.0 benchmark of non-revenue water below 20 % for towns.

13 Hard questions, and our answers

Question	Answer
How can LoRa carry the vibration data?	It does not. Raw bursts (64 kB) stay inside the pair on ESP-NOW; LoRa carries a ~110-byte result. See Part 7.
What limits the ± 1 m accuracy?	Timing and wave speed. Sub-millisecond beacon sync gives ≤ 0.2 m ($v \cdot \delta t / 2$); One-Tap Commissioning removes the wave-speed error. Real dispersion and echoes are measured on the rig.
How do you synchronise two nodes tens of metres apart?	Reference broadcast: both record the arrival time of the same gateway beacon, start a fixed delay later, and a second beacon corrects crystal drift (≤ 40 ppm ≈ 0.6 ms over 15 s before correction).
Why an accelerometer rather than a hydrophone?	Clamp-on at existing valves and hydrants: no pipe cutting, no wet installation. The ADXL355 class has low noise; on plastic pipes the node spacing is shortened.
Why 02:00-04:00?	Lowest legitimate demand and ambient noise. The window is a parameter per village and is also what leak surveyors use.
What if there is real night consumption?	GATE compares against a learned zone baseline, not against zero; only a sustained rise raises the alarm.
How is this different from JJM's flow-based IoT?	Those meters detect loss at zone level. JalNaadi locates it along the pipe and reuses their data as the screening layer.
Is the “AI” real?	The node uses a rule-based spectral classifier; the server adds an Isolation Forest anomaly model; GATE uses EWMA + CUSUM. We make no deep-learning claim.
Battery life?	Calculated ≈ 16 mAh/day $\rightarrow \approx 156$ days on one 2,500 mAh cell; a 1 W panel gives $> 30\times$ margin. Verified on the rig with INA219.
What does it cost?	Indicative node BOM ₹3,000–5,500 (ADXL345 vs ADXL355); the brief itself notes current technologies are expensive for rural use.
What data trains and tests it?	No public dataset exists, so we record our own labelled rig data, add synthetic injection, and request pilot data from State PHED / NJJM.
Metal versus plastic pipes?	Per-segment tap calibration handles speed; spacing is set per material (shorter for plastic).
Is it indigenous?	Design, firmware, analytics and integration are ours; parts are commercial off-the-shelf, assembled in India, with an open-source stack and open data format.

14 Project status — keep every statement truthful

Item	Status
Idea, architecture, equations, power budget, BOM range	Designed and calculated (in the deck and this document)
Localisation, sync sensitivity, one-tap calibration, quantisation, GATE detector, classifier	Simulated in Python; unit tests pass (analysis/)
Backend ranking and ingest	Implemented and tested in software (backend/)
Firmware for EAR / GATE / gateway	Reference skeletons written; not yet run on hardware
Lab rig, measured accuracy, real recordings, field data	To be built and measured by the team

Language rule for the demo and for judges: say *designed*, *calculated* and *simulated* for what is, and *measured* only for what the rig has actually shown. That wording is both honest and credible — and it makes the prototype results, when they arrive, the strongest part of the story.

Appendix A — Glossary

Term	Meaning
Accelerometer	A chip that measures vibration (acceleration); here clamped to a pipe
Beacon	A short radio packet from the gateway used as a shared time reference
CUSUM	Cumulative-sum change detector: adds up small persistent deviations until they cross a threshold
EWMA	Exponentially weighted moving average — a baseline that follows slow drift
ESP-NOW	Short-range peer-to-peer radio built into the ESP32, ~250 bytes per packet
GCC-PHAT	Generalised cross-correlation with phase transform: finds the delay between two signals using only their phase, robust to coloured noise
LoRa	Long-range low-power radio; licence-exempt at 865–867 MHz in India
MQTT / TLS	Lightweight messaging protocol / encryption for the connection
Night flow	Flow into a zone in the quiet hours; a rise above baseline suggests leakage
NRW	Non-revenue water
TDOA	Time difference of arrival
One-Tap Commissioning	Tapping the pipe at install time so the nodes measure the wave speed $v = L/\Delta t$
Sujal Gaon / Sujalam Bharat	JJM 2.0 service-delivery framework and its village-level IDs/dashboard

Appendix B — Official and technical references

Source	Link
MoJS · NJJM Technical Expert Committee Report on measurement and monitoring (Mar 2021)	jaljeewanmission.gov.in/sites/default/files/manual_document/technical-expert-committee-report-on-measuring-and-monitoring_0.pdf
PIB · JJM 2.0 approval and status (31 Mar 2026)	pib.gov.in/PressReleasePage.aspx?PRID=2247224
PIB · JJM sensor-based IoT devices (31 Mar 2021)	pib.gov.in/PressReleasePage.aspx?PRID=1708701
PIB · NJJM measurement and monitoring (16 May 2021)	pib.gov.in/PressReleaselframePage.aspx?PRID=1719104
JJM national dashboard	ejalshakti.gov.in/jjmreport/JJMIndia.aspx
MoHUA · AMRUT 2.0 Operational Guidelines (NRW below 20 %)	mohua.gov.in/upload/whatsnew/61792765b5387AMRUT-Operational-Guidelines.pdf
DoT / WPC · 865–867 MHz delicensed band, GSR 564(E)	dot.gov.in/sites/default/files/Clarification%20ETA%20865-867%20MHz.pdf
CSIR-CSIO · JJM IoT pilot EoI	csio.res.in/upload/EoI/EoI-JJM.pdf
Datasheets / docs	docs.espressif.com/projects/esp-idf · semtech.com/lora · analog.com/en/products/adxl355.html

Appendix C — Repository map

Folder	Contents
README.md	Overview, status table, quick start
analysis/	DSP (GCC-PHAT, classifier), night-flow detector, priority ranking, simulations, 13 tests
backend/	FastAPI + SQLite ingest, leak placement and ranking, tests
dashboard/	Leaflet map page
firmware/	EAR, GATE and gateway reference skeletons (not yet hardware-verified)
hardware/	BOM, wiring
docs/	This explainer, diagrams, simulation figures and result files